

Kronecker-structured adaptation of random Fourier features for robust nonlinear active noise control

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Abstract

Nonlinear active noise control requires expressive controllers that establish attenuation rapidly under non-Gaussian disturbances. Random Fourier features provide a fixed nonlinear representation, while their coefficient parameterization remains a design variable for adaptation. This paper studies a Kronecker-structured random Fourier feature controller with normalized maximum-correntropy updates. Short factor pairs represent the coefficient vector and adapt through secondary-path-filtered features. The formulation separates feature dimension, factor count, and correntropy width, and gives explicit storage, arithmetic, and factor-increment bounds. At 500 fixed features, a four-pair factorization stores 180 adaptive coefficients, a 64% reduction from the full vector. Under logistic-chaotic excitation, the structured controller reaches approximately -20 dB near 2,000 iterations, compared with approximately -13 dB for the full correntropy controller at the same point; the late-stage separation is about 1 dB. Under α -stable impulsive excitation, both correntropy controllers retain attenuation and the structured controller enters the attenuating regime sooner. Under Gaussian excitation with measurement noise, the structured controller reaches -20 dB roughly 2,900 iterations earlier and finishes about 0.6 dB lower in average noise reduction (ANR). These results position coefficient structure as a practical design variable for initial attenuation speed and adaptive model size.

Keywords: nonlinear active noise control, random Fourier features, Kronecker decomposition, maximum correntropy criterion, normalized adaptive filtering

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1. Introduction

Nonlinear active noise control (ANC) requires a controller that represents nonlinear reference-to-disturbance relationships and learns them quickly enough to establish useful attenuation. Polynomial, functional-link, spline, neural-network, and kernel representations address different forms of nonlinearity [1]. Once a representation is chosen, its coefficient parameterization and update rule determine how the controller approaches an attenuating operating point. The present work examines this adaptation problem in a fixed random-feature space.

Random Fourier features (RFFs) approximate a shift-invariant kernel with an explicit collection of nonlinear features [2]. Their predetermined dimension avoids the expanding dictionary of an unrestricted kernel adaptive filter. Shen et al. [3] developed random-feature approximations for multi-kernel adaptive filtering. For ANC, Deb et al. [4] formulated RFF filtered-x least-mean-square (FxLMS) controllers and extended them to multichannel narrowband control. These developments establish the fixed nonlinear map as a practical representation. They also leave a distinct design variable: how its output coefficients are parameterized and adapted.

Robust adaptation addresses the influence of large residuals. The maximum correntropy criterion (MCC) weights each error by a Gaussian kernel of its magnitude relative to the kernel width, suppressing the influence of sufficiently large excursions [5]. Huang et al. [6] studied variable-width MCC adaptation. Recent ANC work in *Signal Processing* includes Euclidean-direction-search MCC updates [7], projection FxLMS methods for impulsive environments [8], and fractional-order generalized complex correntropy [9]. Xiao et al. [10] combined RFFs with a generalized hyperbolic tangent criterion and conjugate-gradient optimization. These methods act on the loss or adaptation strategy; the present study focuses on a structured parameterization of the feature-space coefficients.

Feature design and robust adaptation have also been combined directly. Ye et al. [11] introduced an enhanced multiple-RFF ANC model that fuses feature information through a projection matrix and uses generalized correntropy. Their approach changes how multiple feature maps are combined. Here, one fixed map is retained and its long output-coefficient vector is represented by pairs of shorter factors. This coefficient-space construction separates the nonlinear dictionary from the number and organization of the adaptive parameters. A related Signal Processing study accelerates nonlin-

ear ANC by organizing functional-link features into a delayless multiband subband structure and deriving stability and complexity results [12]. The present construction addresses a different axis: coefficient structure inside a single fixed RFF map.

Kronecker decompositions provide the required factorization. Related nonlinear ANC uses include low-rank spline adaptive filters [13] and tap-decomposed robust Volterra filters [14]. Published KPD work also develops robust adaptive filters and filtered-x ANC algorithms [15], [16], and [17]. These works apply Kronecker decomposition to linear control-filter estimation, including filtered-x and robust correntropy variants. The present paper instead structures the output coefficients of a fixed random Fourier feature map inside the nonlinear ANC loop, and its experimental target is the transient entry into the attenuating regime. A Kronecker factorization represents the long RFF coefficient vector by pairs of shorter factors. We apply normalized correntropy weighting to the resulting factor regressors and examine initial and late-stage attenuation under three excitation conditions. Its experimental focus is the time required to enter a low-residual operating region, considered together with adaptive coefficient storage and per-iteration arithmetic.

The contributions are as follows:

1. A normalized correntropy recursion is formulated for a Kronecker-structured RFF controller, with explicit projected regressors and factor-update timing (Section 3).
2. Adaptive coefficient storage and arithmetic are expressed in terms of the feature dimension and factor count, and each normalized factor increment is bounded (Section 4).
3. Logistic-chaotic, α -stable impulsive, and Gaussian comparisons separate initial attenuation from the late-stage residual. The chaotic comparison shows an approximately 7 dB separation near 2,000 iterations, and the Gaussian comparison shows an earlier -20 dB crossing by roughly 2,900 iterations (Section 5).

Section 2 defines the ANC model, feature map, and correntropy baseline. Sections 3 and 4 develop the structured recursion and its properties. Section 5 presents the comparisons, and Section 6 concludes the paper.

2. Nonlinear ANC model and random Fourier features

2.1. Signal model

Let the reference vector be

$$\mathbf{x}_n = [x(n), x(n-1), \dots, x(n-M+1)]^T \in \mathbb{R}^M. \quad (1)$$

The primary disturbance at the error location is denoted by $d(n)$. The controller generates $y(n)$, and the secondary path produces

$$y_s(n) = \sum_{\ell=0}^{L_s-1} s_\ell y(n-\ell), \quad e(n) = d(n) - y_s(n) + v(n), \quad (2)$$

where $v(n)$ represents measurement noise when included. The subtraction sign specifies the cancellation convention. The primary and secondary finite impulse response paths used to describe the nonlinear simulation model are

$$P(z) = z^{-3} - 0.3z^{-4} + 0.2z^{-5}, \quad S(z) = z^{-2} + 0.5z^{-3}. \quad (3)$$

The coefficient-vector length of $P(z)$ is 6 and the secondary-path length is $L_s=4$, including leading zeros. A nonlinear primary disturbance is formed from $g(n) = P(z)x(n)$ as

$$d(n) = g(n-2) + 0.08g^2(n-2) - 0.04g^3(n-1). \quad (4)$$

This model combines temporal memory with second- and third-order nonlinear terms. Its linear path responses are illustrated in Fig. 1. The filtered-feature formulation below uses the same secondary-path response as the simulation model, so the reference implementation does not introduce a secondary-path mismatch.

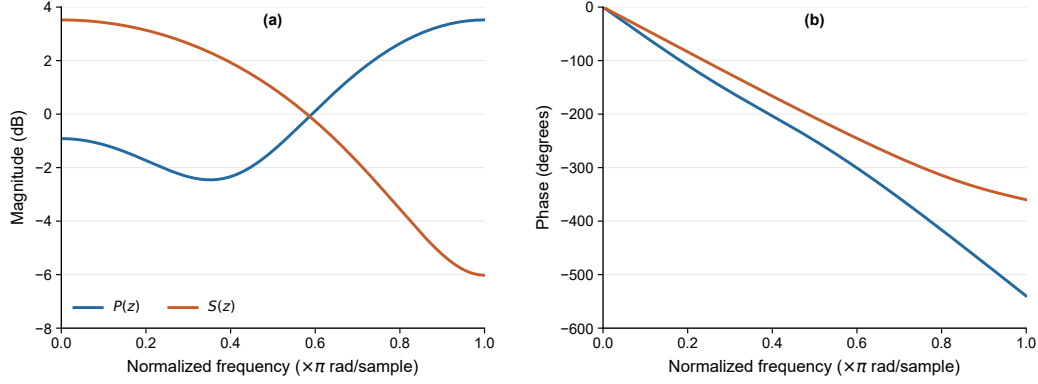


Figure 1: Responses of the primary and secondary linear paths: (a) magnitude and (b) phase. Normalized frequency is expressed relative to π radians per sample.

2.2. Fixed nonlinear feature map

For a Gaussian kernel with length scale ℓ_r , draw independent frequency vectors $\boldsymbol{\omega}_j \sim \mathcal{N}(0, \ell_r^{-2} \mathbf{I}_M)$ and phases $b_j \sim \mathcal{U}[0, 2\pi)$. The identity $\mathbf{I}_M$ acts on the M -dimensional reference vector. The RFF map is

$$\mathbf{z}_n = \sqrt{\frac{2}{D}} \begin{bmatrix} \cos(\boldsymbol{\omega}_1^T \mathbf{x}_n + b_1) \\ \vdots \\ \cos(\boldsymbol{\omega}_D^T \mathbf{x}_n + b_D) \end{bmatrix}. \quad (5)$$

The frequency length scale is $\ell_r = \sigma_{\text{rff}} = 3.9$ in the reference configuration. The frequencies and phases remain fixed during adaptation. Averaging the inner product of two such feature vectors over the random map recovers the Gaussian kernel [2]. A full RFF controller uses

$$y(n) = \mathbf{w}_n^T \mathbf{z}_n, \quad \mathbf{w}_n \in \mathbb{R}^D. \quad (6)$$

The secondary-path-filtered feature vector is

$$\mathbf{q}_n = \sum_{\ell=0}^{L_s-1} s_\ell \mathbf{z}_{n-\ell}. \quad (7)$$

Thus filtering acts along time on each feature sequence. In a frozen-coefficient filtered-feature model, the secondary output is approximated by $\mathbf{w}_n^T \mathbf{q}_n$. This model gives a common regression form for the full and factorized controllers.

2.3. Normalized correntropy baseline

For correntropy width $\sigma_c > 0$, define the scaled instantaneous loss and its derivative as

$$\rho_{\sigma_c}(e) = \sigma_c^2 \left[1 - \exp \left(-\frac{e^2}{2\sigma_c^2} \right) \right], \quad \psi_{\sigma_c}(e) = e \exp \left(-\frac{e^2}{2\sigma_c^2} \right). \quad (8)$$

The scale factor in the loss fixes the learning-rate convention. Around zero, the influence $\psi_{\sigma_c}(e)$ behaves approximately as e ; at large magnitudes it approaches zero. A normalized update for the full coefficient vector is

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \mu \frac{\psi_{\sigma_c}(e_n)}{\varepsilon + \|\mathbf{q}_n\|_2^2} \mathbf{q}_n, \quad (9)$$

where $\mu > 0$ is the step size and $\varepsilon > 0$ prevents division by zero. The full name of this baseline is RFF-FxNLMS-MCC, combining RFFs, filtered-x normalized least-mean-square (FxNLMS) adaptation, and MCC. RFF-MCC is used as its short form below. Setting the exponential weight to one gives the RFF-FxNLMS comparator, abbreviated RFF-NLMS.

3. Kronecker-structured correntropy adaptation

3.1. Coefficient factorization

Choose $D = D_1 D_2$ and represent the coefficient vector as

$$\mathbf{w}_n = \sum_{r=1}^R \mathbf{a}_{r,n} \otimes \mathbf{b}_{r,n}, \quad \mathbf{a}_{r,n} \in \mathbb{R}^{D_1}, \quad \mathbf{b}_{r,n} \in \mathbb{R}^{D_2}. \quad (10)$$

Here R is the number of Kronecker terms. The corresponding coefficient matrix has rank at most R . Define $\mathbf{A}_n = [\mathbf{a}_{1,n}, \dots, \mathbf{a}_{R,n}]$, $\mathbf{B}_n = [\mathbf{b}_{1,n}, \dots, \mathbf{b}_{R,n}]$, and reshape $\mathbf{q}_n$ columnwise into $\mathbf{Q}_n \in \mathbb{R}^{D_2 \times D_1}$. Then

$$\mathbf{w}_n = \text{vec}(\mathbf{B}_n \mathbf{A}_n^T), \quad \hat{d}_n = \sum_{r=1}^R \mathbf{b}_{r,n}^T \mathbf{Q}_n \mathbf{a}_{r,n}. \quad (11)$$

The output before secondary filtering has the same expression with the reshaped unfiltered feature vector in place of $\mathbf{Q}_n$. Consequently, the factorization changes the adaptive coefficient model while preserving the original random features.

3.2. Projected regressors and factor updates

Stack the factors as $\mathbf{a}_n = \text{vec}(\mathbf{A}_n)$ and $\mathbf{b}_n = \text{vec}(\mathbf{B}_n)$. Their projected regressors are

$$\mathbf{u}_{b,n} = \text{vec}(\mathbf{Q}_n \mathbf{A}_n) \in \mathbb{R}^{D_2 R}, \quad \mathbf{u}_{a,n} = \text{vec}(\mathbf{Q}_n^T \mathbf{B}_n) \in \mathbb{R}^{D_1 R}. \quad (12)$$

Both expressions produce the same predicted disturbance:

$$\hat{d}_n = \mathbf{b}_n^T \mathbf{u}_{b,n} = \mathbf{a}_n^T \mathbf{u}_{a,n}. \quad (13)$$

The two normalized factor updates are

$$\begin{aligned} \mathbf{b}_{n+1} &= \mathbf{b}_n + \mu_b \frac{\psi_{\sigma_c}(e_n)}{\varepsilon + \|\mathbf{u}_{b,n}\|_2^2} \mathbf{u}_{b,n}, \\ \mathbf{a}_{n+1} &= \mathbf{a}_n + \mu_a \frac{\psi_{\sigma_c}(e_n)}{\varepsilon + \|\mathbf{u}_{a,n}\|_2^2} \mathbf{u}_{a,n}. \end{aligned} \quad (14)$$

The regressors in (12) are evaluated using the factors at the beginning of iteration n and are retained for both updates. The factor assignments can therefore be executed consecutively without recomputing the second regressor. Each denominator normalizes the entire stacked regressor across the R terms. Under the frozen-coefficient approximation of Section 2.2, (14) is a normalized correntropy-gradient update for the factor variables. Equal factor step sizes, $\mu_a = \mu_b$, provide a simple implementation; separate notation permits their roles to remain explicit.

The resulting controller is denoted NKP-RFF-FxNLMS-MCC, abbreviated NKP-RFF-MCC, where NKP refers to the nearest-Kronecker-product representation motivating (10). During adaptation, the factors are updated directly; an explicit matrix singular-value decomposition is not required at every sample. This makes the factor count a parameter of the adaptive model itself.

For the implementation, the residual used in the correntropy influence is evaluated as

$$e_n = d(n) - \mathbf{b}_n^T \mathbf{u}_{b,n} + v(n), \quad (15)$$

with the measurement-noise term omitted in the noiseless conditions. This is the frozen-coefficient residual used for both factor updates at iteration n .

3.3. Initialization and implementation

The full RFF coefficient vector can be initialized at zero. The factorized controller requires nonzero factors because setting both A and B to zero would also set both regressors in (12) to zero. Small independent Gaussian initial values provide a reproducible initialization, for example 0.01 times standard-normal draws. The random-map seed and factor-initialization seed form part of the experiment configuration.

Algorithm 1. Normalized Kronecker-RFF correntropy adaptation

1. Fix D_1 , D_2 , R , the random frequencies and phases, σ_c , μ_a , μ_b , and ε . Initialize A and B with small nonzero values.
2. Form the reference vector and compute $\mathbf{z}_n$ using (5).
3. Filter the feature sequences through the secondary-path model to obtain $\mathbf{q}_n$, and reshape it into $\mathbf{Q}_n$.
4. Compute both projected regressors from the current factors using (12).
5. Evaluate the residual in (15), omitting measurement noise when absent, and evaluate the correntropy influence in (8).
6. Update b and a using the retained regressors and (14), then reshape them into B and A.
7. Advance the feature and path buffers and repeat.

The scalar correntropy weight and the normalized factor directions perform different tasks. Correntropy regulates the effect of residual magnitude. The Kronecker factors regulate the directions through which the feature-space coefficient model changes. Their combination supplies a structured nonlinear controller with bounded instantaneous error influence.

4. Structural and computational analysis

4.1. Adaptive coefficient storage

The unrestricted model stores D adaptive coefficients. The Kronecker representation stores $R(D_1 + D_2)$, giving the reduction fraction

$$\eta_{\text{store}} = 1 - \frac{R(D_1 + D_2)}{D_1 D_2}. \quad (16)$$

Table 1: Adaptive coefficient counts for representative factorizations.

D	$D_1 \times D_2$	R	Full coefficients	Factor coefficients	Reduction
200	20×10	4	200	120	40%
200	20×10	6	200	180	10%
500	25×20	4	500	180	64%
500	25×20	6	500	270	46%

Storage is reduced when $R < D_1 D_2 / (D_1 + D_2)$. For a fixed product $D_1 D_2$, near-balanced factor dimensions reduce their sum. The RFF frequency matrix and phase vector are fixed parameters shared by both representations and are additional to these adaptive-coefficient counts.

These counts establish how the factor number controls adaptive model size. In particular, four pairs of lengths 25 and 20 describe a 500-dimensional coefficient vector using 180 adaptive values.

The comparisons use $R=4$, which retains a 64% coefficient-storage reduction at a moderate per-iteration arithmetic increase (Fig. 2); $R=6$ is an alternative at 46%.

4.2. Arithmetic cost

The arithmetic comparison counts dense real multiplications in the filtered-feature learning loop. Random-feature initialization, signal generation, plotting, and metric evaluation are excluded. Cosines, exponentials, and divisions are counted separately. Common scalar constants are precomputed. This convention applies equally to both controllers.

Both controllers require D cosine evaluations. A literal two-factor implementation uses two correntropy evaluations and two normalized scalar divisions, while the baseline uses one of each. When the factor residuals and widths coincide, the exponential can be reused. Table 2 retains the literal two-evaluation convention.

For $D=500$, $M=20$, $L_s=4$, $D_1=25$, $D_2=20$, and $R=4$, the counts are 14,004 and 17,048 multiplications. The factorized model therefore exchanges additional projected-regressor calculations for a smaller adaptive parameter set. This is the relevant design trade-off: coefficient storage and transient behavior are considered together with the adaptation cost. Increasing R

Table 2: Multiplication counts per iteration in the filtered-feature learning loop.

Operation	RFF-MCC	NKP-RFF-MCC
Random-feature projection and scaling	$DM + D$	$DM + D$
Secondary-path filtering of features	DL_s	DL_s
Two factor-regressor contractions	0	$2RD$
Prediction dot products	D	$R(D_1 + D_2)$
Regressor squared norms	D	$R(D_1 + D_2)$
Vector weight increments	D	$R(D_1 + D_2)$
Scalar correntropy/update calculations	4	8
Total	$DM + DL_s + 4D + 4$	$DM + DL_s + D + 2RD + 3R(D_1 + D_2) + 8$

enlarges both the factor model and its per-sample arithmetic, as illustrated in Fig. 2.

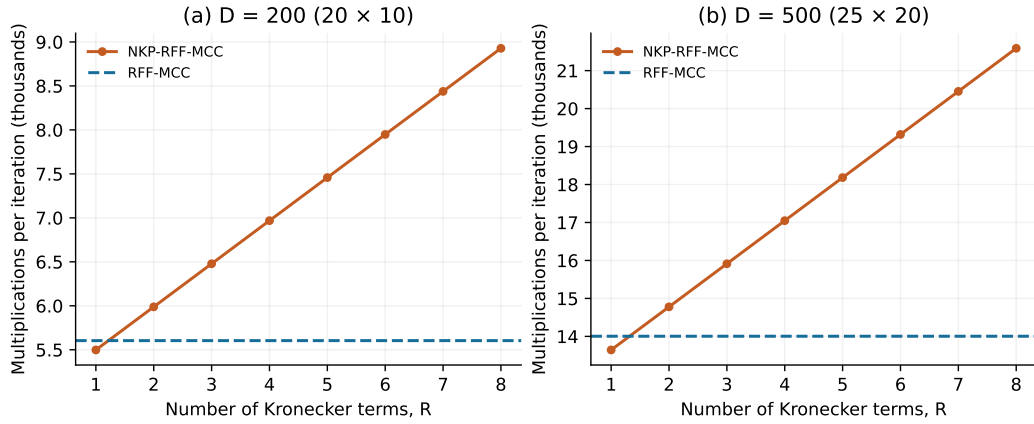


Figure 2: Analytic multiplication counts for the filtered-feature learning loop, with $M=20$ and $L_s=4$. Panels (a) and (b) use $D=200$ and $D=500$, respectively. Solid lines show NKP-RFF-FxNLMS-MCC, and dashed lines show the corresponding full RFF-FxNLMS-MCC counts. Each curve follows Table 2 and includes the random-feature mapping on the same basis.

4.3. Bound on normalized factor increments

The correntropy influence satisfies

$$\sup_e |\psi_{\sigma_c}(e)| = \sigma_c \exp(-1/2). \quad (17)$$

For any regressor $\mathbf{u}$ and $\varepsilon > 0$,

$$\frac{\|\mathbf{u}\|_2}{\varepsilon + \|\mathbf{u}\|_2^2} \leq \frac{1}{2\sqrt{\varepsilon}}. \quad (18)$$

Combining (14), (17), and (18) gives

$$\|\Delta \mathbf{a}_n\|_2 \leq \frac{\mu_a \sigma_c \exp(-1/2)}{2\sqrt{\varepsilon}}, \quad \|\Delta \mathbf{b}_n\|_2 \leq \frac{\mu_b \sigma_c \exp(-1/2)}{2\sqrt{\varepsilon}}. \quad (19)$$

The result bounds individual factor increments for a fixed kernel width. It identifies the direct role of correntropy and normalization in controlling update magnitude, without requiring a distributional assumption on the instantaneous error.

4.4. Interpretation of the factor adaptation

Each projected regressor depends on the companion factor. Thus the same filtered feature vector generates different adaptation directions as $\mathbf{A}$ and $\mathbf{B}$ evolve. A change to one short factor affects a coordinated group of entries in the implied long coefficient vector. This provides a mechanism through which the controller can organize its initial adaptation within the selected factor model.

The factor number R and correntropy width σ_c govern different aspects of this process. R controls the representational structure and coefficient count. The width determines the residual scale emphasized by the update. Their effects should therefore be examined separately: a kernel-parameter sweep addresses error weighting (Section 5.2), while the factor count is selected from the storage–arithmetic trade-off (Section 4, Fig. 2).

5. Simulation results

5.1. Evaluation structure and attenuation measure

The experiments compare RFF-MCC and NKP-RFF-MCC under logistic-chaotic, α -stable impulsive, and Gaussian conditions. RFF-NLMS is included in the impulsive and Gaussian comparisons to show the role

of correntropy weighting. The chaotic and impulsive trajectories cover 20,000 iterations, and the Gaussian comparison covers 30,000 iterations with measurement noise included in the error signal.

Average noise reduction (ANR) is expressed using exponentially smoothed absolute amplitudes,

$$\begin{aligned} A_d(n) &= \beta A_d(n-1) + (1-\beta)|d(n)|, \\ A_e(n) &= \beta A_e(n-1) + (1-\beta)|e(n)|, \\ \text{ANR}(n) &= 20 \log_{10} \frac{A_e(n) + \epsilon_a}{A_d(n) + \epsilon_a}. \end{aligned} \tag{20}$$

Negative values denote attenuation. The reference configuration uses $\beta=0.999$, 20 Monte Carlo runs, and random seed 871. Table 3 specifies the reference configuration and the displayed evaluation intervals. The early learning period and the late residual level are evaluated separately. To summarize the transient, define a first threshold-crossing index

$$t_\tau = \min\{n : \text{ANR}(n) \leq \tau\}.. \tag{21}$$

Numerical summaries in this section are rounded readings of the displayed ANR trajectories. The common model notation is listed in Table 3; the coefficient-count examples retain $D=500$ with a 25×20 factorization and $R=4$.

5.2. Sensitivity to correntropy weighting

The correntropy parameter determines which errors contribute strongly to the update. It is useful to express the sweep using

$$g_k(e) = \exp\left(-\frac{ke^2}{2\sigma_0^2}\right), \quad \sigma_{c,\text{eff}} = \frac{\sigma_0}{\sqrt{k}}. \tag{22}$$

With $\sigma_0=2.0$, $k=1$ gives the comparison width $\sigma_c=2$, and increasing k narrows the effective correntropy kernel. This k is distinct from the factor count R . Fig. 3 compares the RFF-FxNLMS-MCC learning trajectories over the displayed parameter range, and Fig. 4 summarizes their steady-state levels over the displayed final window.

Table 3: Model quantities and evaluation intervals.

Quantity	Definition or setting
M	Reference-vector length; 20
D	500 fixed random features
D_1, D_2	25 and 20, respectively; $D = D_1 D_2$
R	Four Kronecker terms; $R=6$ as an alternative model size
ℓ_r	RFF kernel length scale, 3.9
σ_c	Comparison-family correntropy width, 2
Step sizes	$\mu=0.05$ (baseline, (9)); $\mu_a = \mu_b=0.08$ (factors, (14)) for the logistic-chaotic and α -stable families; $\mu=0.30$; $\mu_a = \mu_b=0.12$ for the Gaussian family
Factor initialization	Independent $0.01 \times$ standard-normal values
Logistic initialization	First six states 0.37; centered and normalized to unit sample standard deviation
α -stable generator	Symmetric CMS generator, $\alpha=1.6$, skewness 0, scale 1, location 0; median-amplitude normalization and clipping to ± 5
Gaussian reference	Standard normal; measurement noise enters the error signal
Reference run protocol	20 Monte Carlo runs; seed 871
$P(z), S(z)$	Primary and secondary paths in (3)
Primary nonlinearity	Delayed polynomial in (4)
β	Absolute-amplitude smoothing factor, 0.999
ε	Update-denominator regularizer in (9) and (14), 10^{-8}
ε_a	ANR regularizer, 10^{-12}
Chaotic and impulsive intervals	20,000 iterations
Gaussian interval	30,000 iterations
Transient thresholds	-20 dB for chaotic/Gaussian; -2 dB for impulsive

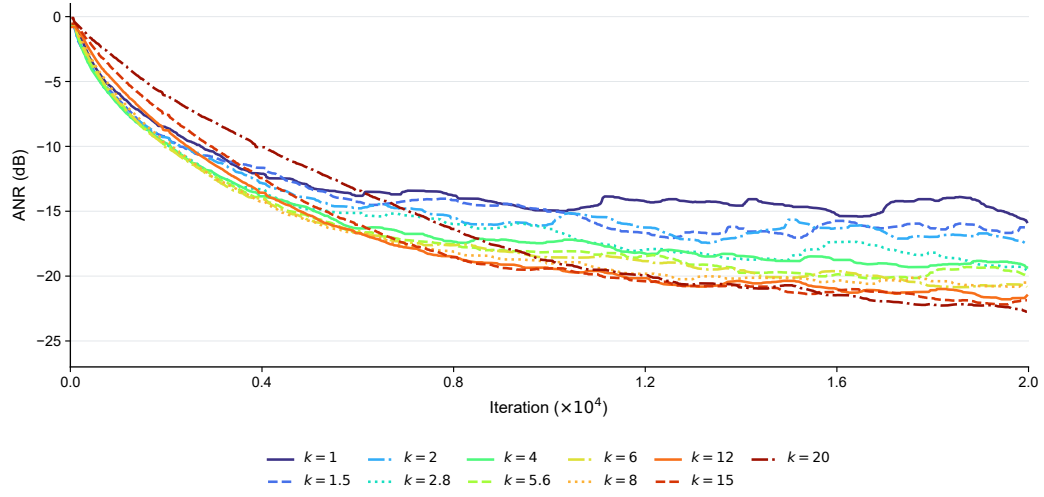


Figure 3: ANR trajectories for the RFF-FxNLMS-MCC correntropy-parameter sweep under logistic-chaotic excitation. The swept values are $k \in 1, 1.5, 2, 2.8, 4, 5.6, 6, 8, 12, 15, 20$; larger k corresponds to stronger suppression of errors outside the effective kernel width in (22).

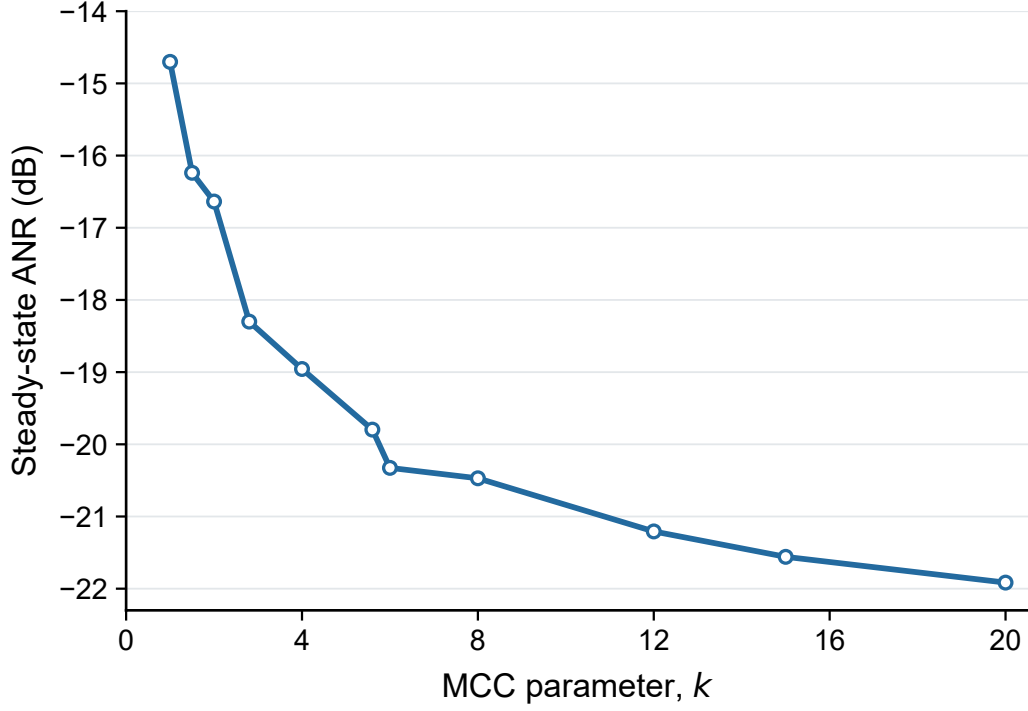


Figure 4: Steady-state ANR over the displayed final window versus the correntropy parameter k . The attenuation changes from approximately -15 dB to approximately -22 dB across the displayed range, with smaller additional changes at larger k .

The curves show that steady-state ANR decreases (deeper attenuation) as k increases. The summary changes by approximately 7 dB across the sweep, and the incremental improvement becomes smaller toward the upper end of the range. This pattern supports treating the error-weighting scale as an important tuning variable. A narrower effective kernel reduces the contribution of larger residual excursions, while retaining updates driven by errors closer to zero. The useful operating range is therefore determined by the resulting attenuation trajectory, rather than by identifying k with the structural factor count.

5.3. Logistic-chaotic reference

The chaotic reference provides a nonlinear, non-Gaussian excitation for comparing the initial learning behavior of the two correntropy controllers. The supporting generator uses a delayed logistic recursion $u(n) = 4u(n -$

$6)[1 - u(n - 6)]$, followed by centering and amplitude normalization. Fig. 5 shows the ANR comparison over 20,000 iterations.

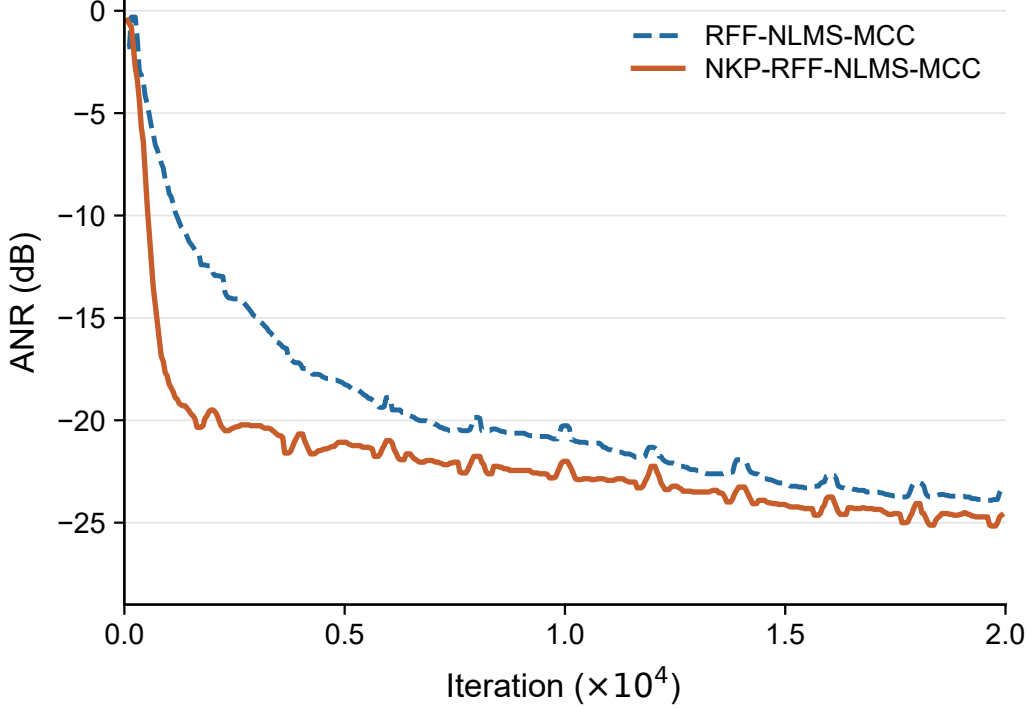


Figure 5: ANR under logistic-chaotic excitation for RFF-FxNLMS-MCC and NKP-RFF-FxNLMS-MCC. The structured controller reaches the -20 dB region earlier and maintains a lower late-stage curve over the displayed interval.

The early learning period shows a clear separation. Near 2,000 iterations, NKP-RFF-MCC is approximately -20 dB, while RFF-MCC is approximately -13 dB. In the plotted comparison, this 7 dB separation coincides with earlier entry into the selected attenuation region by the structured controller. The baseline approaches -20 dB around 6,500–7,000 iterations, whereas the structured curve first touches this region around 1,600–1,700 iterations. Thus the first threshold crossing occurs roughly 5,000 iterations earlier in this comparison.

The curves subsequently approach one another as both controllers continue adapting. At 20,000 iterations, their levels are approximately -24 dB and -25 dB for the baseline and structured controller, respectively. The approximately 1 dB separation at the end complements the more pronounced

transient improvement. For this condition, the principal benefit is earlier access to a low-residual operating region.

5.4. Impulsive-noise condition

The α -stable comparison examines adaptation when large noise excursions occur. Fig. 6 includes the non-correntropy RFF-NLMS controller alongside the two MCC controllers. This separates the effect of robust error weighting from the transient behavior associated with coefficient structure.

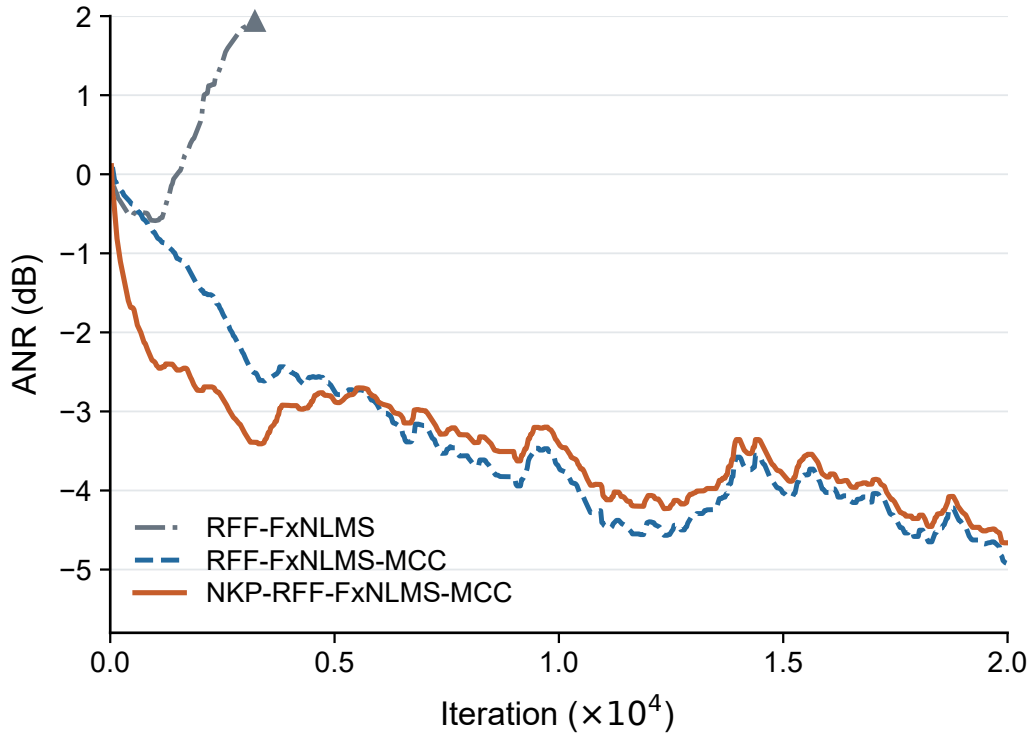


Figure 6: ANR in the α -stable impulsive-noise comparison. RFF-FxNLMS rises above the plotted +2 dB range. Both correntropy controllers maintain negative ANR over the later interval, and NKP-RFF-FxNLMS-MCC reaches -2 dB earlier.

RFF-NLMS initially reduces the residual, then rises above 0 dB and exits the displayed range at +2 dB. The correntropy controllers instead maintain attenuation throughout the later portion of the experiment. This difference is consistent with the error weighting in (8): large residuals generate decreasing influence once their magnitude exceeds the kernel scale.

The structured controller first crosses -2 dB within approximately 600–800 iterations, compared with 2,600–2,900 iterations for the full correntropy controller. Its early trajectory therefore establishes attenuation sooner in the impulsive condition as well. At the end of the 20,000-iteration interval, the full and structured correntropy curves are approximately -4.9 dB and -4.7 dB. The two methods reach similar late-stage levels after different initial transients. The experimentally supported distinction is the faster onset of attenuation together with the continued attenuation supplied by correntropy weighting.

5.5. Gaussian reference with measurement noise

The Gaussian comparison evaluates behavior under a conventional reference distribution with measurement noise in the error signal. Fig. 7 shows that all three controllers approach a stable residual level. The two unrestricted RFF controllers follow closely spaced trajectories, while the structured controller descends more rapidly during the initial learning period.

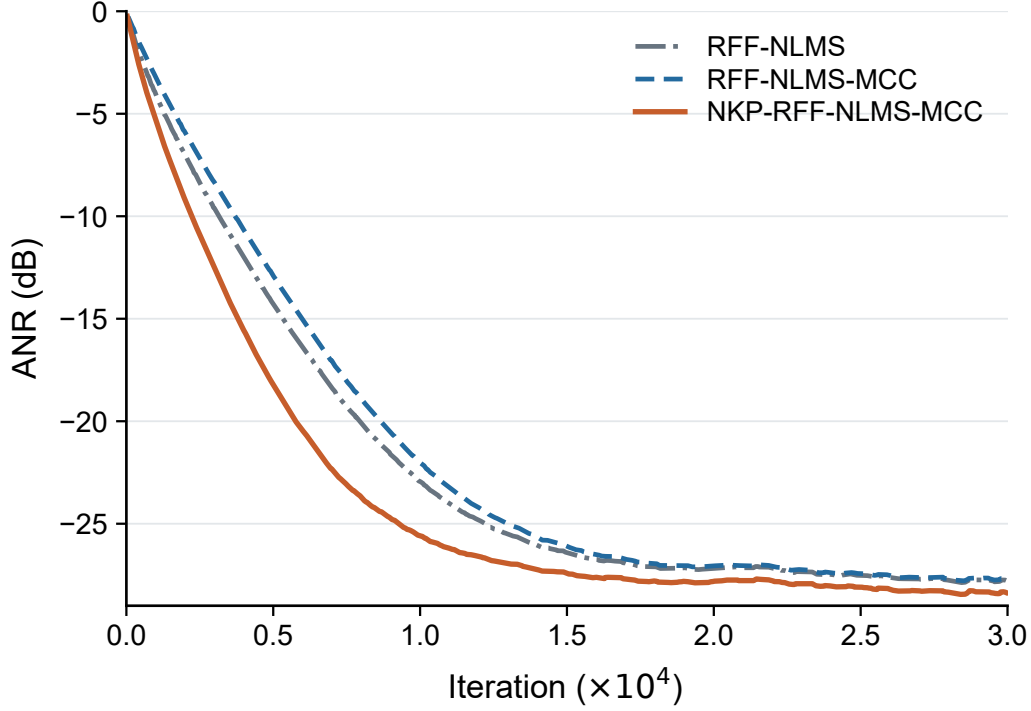


Figure 7: ANR for a Gaussian reference with error-signal measurement noise over 30,000 iterations. NKP-RFF-FxNLMS-MCC reaches the -20 dB region earlier; the three curves approach closely spaced late-stage levels.

NKP-RFF-MCC crosses -20 dB within approximately 5,700–6,000 iterations. The full correntropy controller reaches the same level within 8,500–8,900 iterations, and RFF-NLMS does so near 8,000 iterations. The structured-versus-MCC separation is therefore about 2,500–3,200 iterations, with a central reading near 2,900. Where impulsive excursions are absent, correntropy weighting mildly slows the unstructured controller; the structured controller nonetheless attains the earliest crossing.

By 30,000 iterations, the full controllers lie near -27.8 dB and the structured controller near -28.4 dB. The late-stage separation is approximately 0.6 dB. The main result is again the earlier attenuation, accompanied here by a modest lower final residual. The comparison also shows that the transient benefit remains visible when impulsive excursions are not the dominant feature of the reference signal.

Table 4: Approximate first threshold crossings read from the ANR curves.

Condition	Threshold	RFF- FxNLMS- MCC	NKP-RFF- FxNLMS- MCC	Earlier crossing
Logistic- chaotic	−20 dB	6,500–7,000	1,600–1,700	About 5,000 iterations
α -stable impulsive	−2 dB	2,600–2,900	600–800	About 1,800–2,300 iterations
Gaussian with measurement noise	−20 dB	8,500–8,900	5,700–6,000	About 2,500–3,200 iterations

5.6. Comparison across conditions

Table 4 collects the transient readings from the three experiments. The thresholds differ across conditions because their attainable attenuation levels differ. The ranges are read from the digitized curves using the first-touch criterion for every curve and do not represent between-run statistical intervals. Each comparison uses the same threshold for the two correntropy controllers within its condition.

The consistent observation is an earlier approach to the selected attenuation level. The chaotic and Gaussian comparisons additionally retain lower late-stage curves for the structured controller. In the impulsive comparison, the two correntropy controllers achieve nearby final levels while the non-correntropy controller leaves the attenuating regime. These results distinguish three roles within the method: the RFF map supplies nonlinear features, the Kronecker model organizes coefficient adaptation, and correntropy moderates the response to large errors. The evidence supports a transient-speed and model-size claim.

6. Conclusion

A Kronecker-structured RFF controller with normalized correntropy adaptation has been formulated for nonlinear ANC. Its factor updates operate on secondary-path-filtered features, separating the nonlinear dictionary

from the adaptive coefficient structure. A four-term 25×20 factorization reduces the adaptive coefficient count from 500 to 180, while explicit arithmetic counts quantify the associated factor-regressor computations.

The experiments emphasize faster initial attenuation. Under logistic-chaotic excitation, the structured controller is approximately 7 dB lower near 2,000 iterations and retains an approximately 1 dB late-stage advantage. In the Gaussian comparison, it reaches -20 dB roughly 2,900 iterations earlier. Under impulsive excitation, correntropy maintains attenuation and the structured model enters the attenuating region sooner. Together, these results identify structured coefficient adaptation as a useful design choice when the speed of initial noise reduction matters alongside adaptive model size. The next high-value validation is a controlled secondary-path mismatch study with the same factor and feature budgets.

CRedit authorship contribution statement

[Author-specific contributions to be supplied: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Supervision, and Funding acquisition, as applicable.]

Declaration of competing interest

[Competing-interest declaration to be completed by the authors.]

Data availability

[Data and code availability statement to be completed by the authors.]

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